



Technical Report Guideline — SPY Market Regime Research



Core writing rule: viết report như một mini research paper, không chia chapter theo role và không kể theo thứ tự code. Mỗi phần nên đi theo logic **Question** → **Method** → **Evidence** → **Interpretation**.

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I. Abstract

Mục tiêu: giúp người đọc hiểu toàn bộ nghiên cứu trong khoảng 1 trang.

Nên viết theo thứ tự:

1. **Problem:** ML signal trong financial markets có thể thay đổi predictive power khi market conditions thay đổi.
2. **Research question:** nêu câu hỏi nghiên cứu chính.
3. **Experimental setup:** SPY daily data, 5-trading-day forward return, LowVol/HighVol, purged walk-forward, zero baseline, LightGBM và các model so sánh.
4. **Main findings:** Các kết quả cần đạt
5. **Conclusion:** 1–2 câu trả lời research question.

Độ dài gợi ý: 0.5–1 trang.

II. Introduction and Research Questions

II.1 Background and Motivation

Giải thích ngắn:

- Short-horizon return prediction là bài toán low signal-to-noise.
- Financial relationships có thể không stationary.
- Aggregate OOS performance có thể che giấu sự khác biệt giữa các market regimes.
- Volatility là một regime dimension đơn giản và phù hợp với scope project.

Kết phần này bằng vấn đề:

Một model hoạt động overall không có nghĩa predictive information của nó giữ nguyên trong mọi volatility regime.

II.2 Research Problem

Formulate bài toán:

$$X_t \rightarrow \hat{r}_{t,t+5}$$

với target:

$$r_{t,t+5} = \frac{P_{t+5}}{P_t} - 1$$

Nhấn mạnh project nghiên cứu **predictive information**, không phải trading profitability.

II.3 Research Questions

RQ1 — Predictive Value

Does the primary LightGBM signal provide out-of-sample predictive value relative to a zero-return baseline?

RQ2 — Regime Dependence

Does the out-of-sample predictive performance of the LightGBM signal differ between Low-Vol and High-Vol market regimes?

RQ3 — Model Robustness

Are the observed predictive and regime-dependent patterns specific to LightGBM, or do they persist across alternative tree-ensemble methods?

RQ3 là nơi hợp lý để đưa Random Forest, XGBoost và AdaBoost vào report mà không biến toàn bộ project thành bài toán “model nào tốt nhất”.

II.4 Hypothesis

Cho RQ2:

$$H_0 : Performance_{LowVol} = Performance_{HighVol}$$

$$H_1 : Performance_{LowVol} \neq Performance_{HighVol}$$

Nếu project chưa chạy formal significance test, nói rõ đây là **research hypothesis**, còn evidence chính hiện tại là comparative OOS metrics.

II.5 Contributions

Nên nêu 5–6 đóng góp:

- Point-in-time SPY modeling dataset.
- Historical-only volatility regime.
- Leakage-safe 5-day purged walk-forward validation.
- Genuine OOS prediction framework.
- Zero-baseline-controlled evaluation.
- Cross-model robustness comparison.
- Reproducible and auditable pipeline.

III. Related Work and Research Gap

Độ dài gợi ý: 2–3 trang.

III.1 Machine Learning for Return Prediction

Review ngắn:

- Statistical/linear forecasting.
- Tree ensembles.
- Gradient boosting.
- Challenges of short-horizon return prediction.

Mục tiêu là dẫn đến câu hỏi: **tại sao nonlinear tabular models có thể phù hợp nhưng vẫn dễ overfit noise?**

III.2 Gradient Boosting and Tree Ensembles

Review LightGBM, XGBoost, AdaBoost và Random Forest ở mức liên quan tới experiment:

- boosting vs bagging;
- bias/variance;
- nonlinear interactions;
- regularization;
- noise sensitivity.

Không cần viết dài về từng algorithm, chỉ cần giữ ở mức literature review.

III.3 Market Regimes and Non-Stationarity

Review:

- regime-dependent predictive relationships;
- volatility states;
- model instability khi distribution thay đổi.

Không mở rộng sang HMM/macro regimes nếu project không sử dụng.

III.4 Validation in Financial Machine Learning

Tập trung:

- look-ahead bias;
- temporal ordering;
- overlapping forward labels;
- walk-forward evaluation;
- genuine OOS predictions.

III.5 Research Gap

Đoạn này phải dẫn thẳng tới project:

Aggregate OOS performance alone does not reveal whether the same learned signal behaves consistently under different volatility conditions. Overlapping forward-return labels also require explicit temporal controls to avoid leakage.

Kết bằng cách project xử lý gap:

- point-in-time features;
- historical regime threshold;
- purged walk-forward;
- same-row OOS comparison.

IV. Methodology and Research Architecture

Mục tiêu: định nghĩa experiment, chưa trình bày actual numerical results.

IV.1 Problem Formulation

Define:

$$\hat{y}_t = f(X_t)$$
$$y_t = \frac{Close_{t+5}}{Close_t} - 1$$

Giải thích:

- X_t : information available no later than t .
- y_t : realized future return.
- $\hat{y}_t$: ML predictive signal.

IV.2 Baseline

$$\hat{y}_t^{baseline} = 0$$

Rationale:

- short-horizon equity returns noisy và thường gần zero;
- zero predictor là benchmark đơn giản nhưng meaningful theo MAE.

IV.3 Primary Model — LightGBM

Giải thích lý do chọn **a priori**:

- phù hợp tabular numerical features;
- học nonlinear relationships/interactions;
- efficient repeated fitting;
- representative gradient-boosting implementation.

LightGBM is the primary research model by experimental design, not because it necessarily achieves the best OOS result.

IV.4 Comparative Models

Random Forest

Vai trò: bagging / variance-reduction benchmark.

XGBoost

Vai trò: alternative regularized gradient-boosting implementation.

AdaBoost

Vai trò: classical boosting benchmark.

Model	Family	Reason included	Main trade-off
LightGBM	Gradient Boosting	Primary research model	Can fit noise aggressively
XGBoost	Gradient Boosting	Alternative regularized GB	More complexity
AdaBoost	Boosting	Classical benchmark	Noise sensitivity
Random Forest	Bagging	Variance-reduction control	Less residual adaptation

IV.5 Market Regime Definition

Define 20-day realized volatility:

$$RV_{20,t} = std(r_{t-19}, \dots, r_t)$$

Historical threshold:

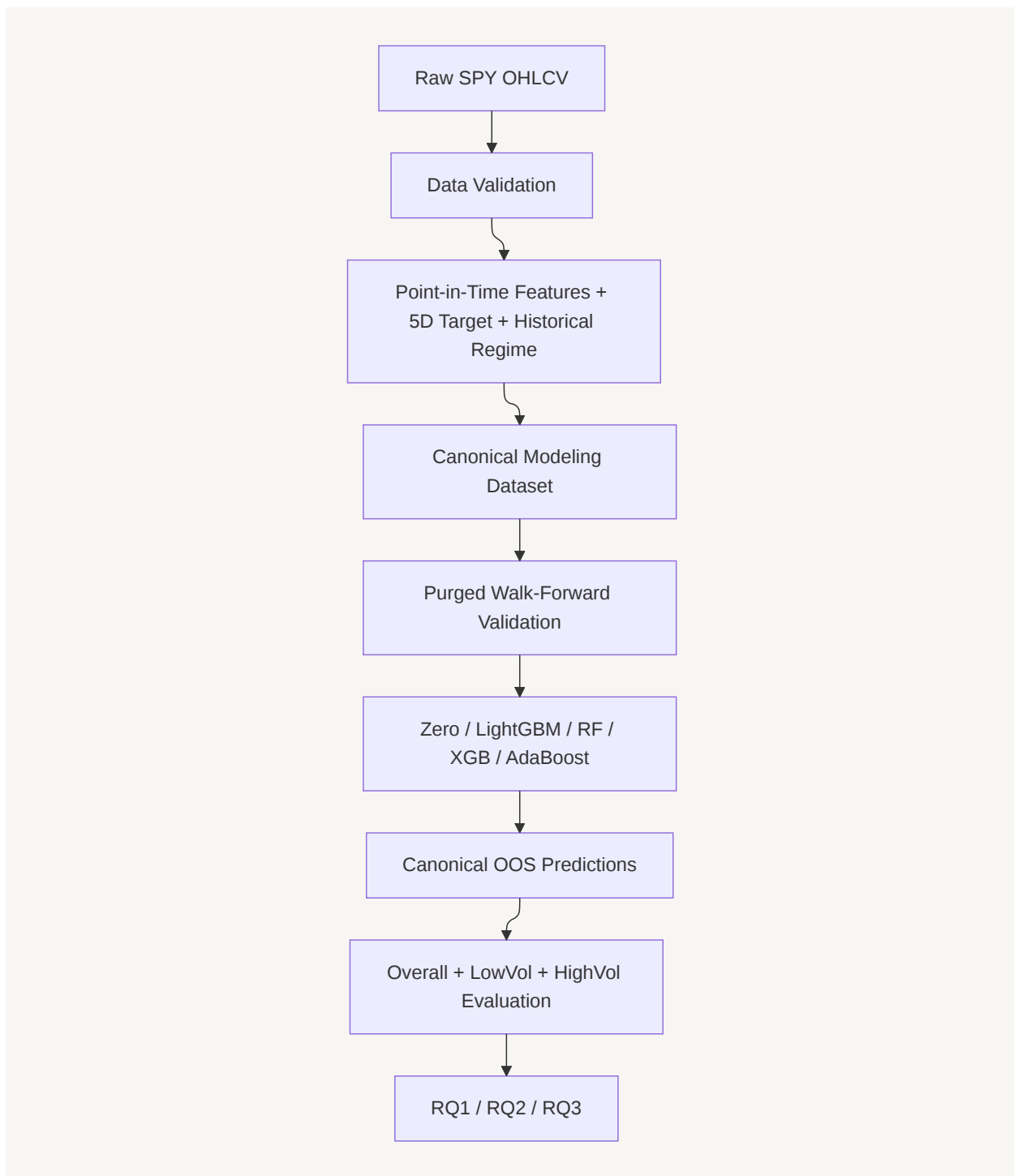
$$c_t = Median(RV_{20,1}, \dots, RV_{20,t-1})$$

Regime:

$$Regime_t = \begin{cases} LowVol, & RV_{20,t} < c_t \\ HighVol, & RV_{20,t} \geq c_t \end{cases}$$

Phải giải thích tại sao **không dùng full-sample median**: threshold tại quá khứ sẽ chứa information về future volatility distribution.

IV.6 Research Architecture



V. Data and Feature Engineering

V.1 Data Source

Nêu:

- Asset: SPY.
- Frequency: Daily.
- Source: Yahoo Finance via `yfinance`.
- OHLCV.
- Adjusted-price convention.
- Date range.

Không paste download code.

V.2 Data Validation

Summary các checks:

- chronological index;
- duplicate dates;
- missing OHLCV;
- positive prices;
- OHLC consistency;
- volume validity;
- calendar-gap sanity check.

Validation	Result
Missing OHLCV	...
Duplicate dates	...
Material OHLC violations	...
Negative volume	...
Pre-model leakage/data audit	PASS / BLOCKED

Dùng ngôn ngữ general như:

| No material data-quality violation was identified under the defined validation rules.

V.3 Target Construction

$$forward_return_{5d,t} = \frac{Close_{t+5}}{Close_t} - 1$$

Trong phần này, tập trung giải thích công thức, chú ý đến

- 5 = five trading observations;
- target aligned at prediction date t ;
- future price chỉ dùng trong label;
- final five observations không có valid target.

V.4 Feature Engineering

Feature set hiện tại gồm 11 features.

Group	Features	Purpose	Information cutoff
Return	1D, 5D, 10D, 20D	Recent price movement	$\leq t$
Volatility	5D, 10D, 20D	Recent variability	$\leq t$
Trend	10D, 20D, 60D	Price relative to trend	$\leq t$
Volume	20D ratio	Relative activity	$\leq t$

Nguyên tắc:

$$X_t = f(\mathcal{I}_t)$$

trong đó $\mathcal{I}_t$ chỉ chứa information available at or before t .

V.5 Canonical Modeling Dataset

Phải report:

- final date range;
- final sample size;
- feature count;
- regime counts;
- missing/inf policy;
- unique sorted dates.

Critical rule:

Downstream models use the canonical dataset rather than reconstructing features, target or regime independently.

VI. Experimental Protocol and Validation

Đây là phần quan trọng nhất về scientific validity.

VI.1 Controlled Comparison

Fixed factors

Tất cả models dùng cùng:

- canonical dataset;
- feature set;
- target;
- regime labels;
- fold boundaries;
- 5D purge;

- OOS dates;
- metric implementation.

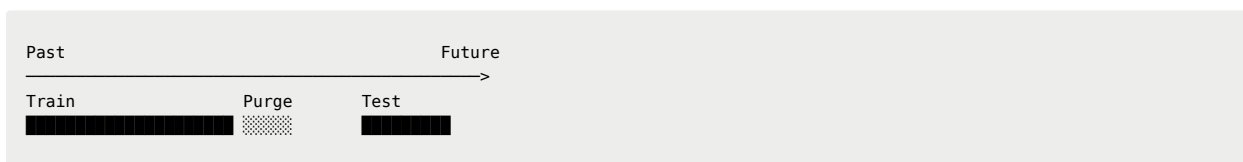
Changed factor

Model Algorithm

VI.2 Walk-Forward Validation

Phải nêu:

- chronological split;
- expanding training window;
- no shuffle;
- genuine future test blocks;
- six folds nếu final configuration vẫn giữ như repo hiện tại.



VI.3 Five-Day Purge

Target:

$$y_t = \frac{P_{t+5}}{P_t} - 1$$

Nếu label của training row gần boundary dùng một future price thuộc test period thì xảy ra leakage.

Do đó:

$$Gap = 5 \text{ trading observations}$$

Nên coi đây là **methodological validity requirement**, không phải implementation preference.

VI.4 OOS Prediction Definition

Một prediction là genuine OOS khi model tạo $\hat{y}_t$ chưa từng fit trên observation t hoặc bất kỳ future information nào thuộc test fold đó.

Canonical OOS schema:

```
Date
prediction
actual_return_5d
regime
fold_id
```

VI.5 Evaluation Metrics

Primary metric — MAE

$$MAE = \frac{1}{N} \sum_i |y_i - \hat{y}_i|$$

Secondary — Prediction-return correlation

$$Corr(\hat{y}, y)$$

Secondary — Directional hit rate

$$P(\text{sign}(\hat{y}) = \text{sign}(y))$$

Luôn report N .

VI.6 Model Ranking Rule

Định nghĩa trước Results:

OOS MAE is the primary ranking metric. Prediction-return correlation and directional hit rate are complementary diagnostics.

Không cherry-pick model dựa trên metric có lợi nhất.

VI.7 QA Gates

Pre-model gate

Kiểm tra:

- point-in-time features;
- target alignment;
- historical regime threshold;
- canonical dataset integrity.

OOS integrity gate

Kiểm tra:

- train < test chronology;
- 5D purge;
- no in-sample predictions;
- no duplicate OOS dates;
- baseline và model comparison dùng identical rows.

VII. Results and Discussion



Rule: Results sections phải map trực tiếp với Research Questions. Không organize Results theo folder/code/module.

VII.1 RQ1 — Overall Predictive Value

Main table:

Model	N	OOS MAE	Improvement vs Zero	Correlation	Hit Rate
Zero				N/A	N/A
LightGBM					

Discussion trả lời:

- LightGBM beat baseline không?
- Correlation có positive không?
- Directional prediction như thế nào?
- Metrics có agree không?

VII.2 RQ2 — LightGBM Across Regimes

Scope	N	Baseline MAE	LightGBM MAE	Improvement	Correlation	Hit Rate
Overall						
LowVol						
HighVol						

Critical interpretation rule:

Không chỉ thấy $MAE_{HighVol} > MAE_{LowVol}$ rồi kết luận model deteriorates. HighVol periods vốn có return magnitude lớn hơn; cần xem cả performance **relative to baseline**, correlation và hit rate.

VII.3 RQ3 — Multi-Model Comparison

Rank	Model	MAE	Δ vs baseline	Correlation	Hit Rate
	Random Forest				
	XGBoost				
	LightGBM				
	AdaBoost				

Report đúng evidence, không điều chỉnh narrative để LightGBM thắng nếu kết quả không support, khi viết cần chú ý các argument, trade-offs để đưa LightGBM lên.

VII.4 Cross-Model Regime Robustness

Model	LowVol MAE	HighVol MAE	LowVol Δ baseline	HighVol Δ baseline
Random Forest				
XGBoost				

Model	LowVol MAE	HighVol MAE	LowVol Δ baseline	HighVol Δ baseline
LightGBM				
AdaBoost				

Question chính:

Regime pattern có chỉ xảy ra với LightGBM hay xuất hiện across ensemble methods?

VII.5 Metric Disagreement

Phân tích khi MAE, correlation và hit rate không cùng hướng.

Ví dụ nếu AdaBoost vẫn có positive correlation nhưng low hit rate thì đây là finding cần report, không phải số liệu phải bỏ.

VII.6 Error Analysis

Chỉ giữ analyses có research value:

- largest OOS errors;
- extreme-return periods;
- regime-specific error distributions;
- fold-level instability;
- nearly constant predictions.

VII.7 Interpretation

Trong phần này, cần tách rõ 2 thứ:

- **Observation:** điều data trực tiếp support.
- **Possible explanation:** interpretation hợp lý nhưng chưa causal.

Ví dụ:

Observation: Random Forest achieved lower OOS MAE.

Possible explanation: averaging decorrelated trees may provide useful variance reduction in this low-signal setting.

Không viết possible explanation như proven cause.

VII.8 Summary of Results

Mỗi RQ đúng một kết luận ngắn:

- **RQ1:** ...
- **RQ2:** ...
- **RQ3:** ...

Không thêm analysis mới tại đây.

VIII. Reliability and Limitations

VIII.1 Internal Validity

Nêu các controls:

- historical-only features;
- historical-only regime threshold;
- purged forward labels;
- chronological split;
- identical OOS comparison rows;
- independent QA gates.

VIII.2 Experimental Limitations

Ví dụ:

- fixed hyperparameters;
- limited candidate models;
- one historical realization;
- overlapping 5D outcomes;
- limited uncertainty analysis nếu chưa thực hiện.

VIII.3 External Validity

Claims chỉ support trực tiếp cho:

- SPY;
- daily data;
- selected historical sample;
- current 11-feature set;
- current binary volatility regime.

Không generalize thành "gradient boosting fails in financial markets".

VIII.4 Interpretation Boundaries

| Predictive performance does not imply trading-strategy profitability.

| Regime-performance association does not establish causality.

Project hiện không bao gồm:

- transaction costs;
- turnover;
- position sizing;

- portfolio construction;
 - execution/risk layer.
-

IX. Conclusion and Future Work

IX.1 Conclusion

Không recap toàn methodology mà trả lời trực tiếp:

1. **RQ1:** LightGBM có cung cấp incremental OOS predictive value hay không?
2. **RQ2:** LightGBM performance có khác giữa LowVol/HighVol hay không?
3. **RQ3:** Model robustness comparison củng cố hay làm yếu conclusion ban đầu?

Sau đó viết một paragraph synthesis.

IX.2 Future Work

Giữ khoảng 4–6 directions:

- multiple assets;
 - alternative regime definitions;
 - richer feature sets;
 - statistical uncertainty/significance;
 - concept drift/adaptive models;
 - strategy-level evaluation with transaction costs.
-

X. References

Cần academic references cho:

- LightGBM;
- XGBoost;
- Random Forest;
- AdaBoost;
- financial ML validation;
- return forecasting;
- regime-dependent modeling;
- data source khi cần.

Không dùng GitHub README/blog tutorial thay cho literature chính.

Appendix

Đẩy engineering details xuống đây:

- full feature data dictionary;
- full hyperparameters;
- six fold boundaries;
- full results by fold;
- full model × regime tables;
- QA checklist/results;
- pipeline config;
- package versions;
- repository tree;
- reproduction commands;
- manifest/hash information;
- additional plots.

Recommended Figures and Tables

Artifact	Purpose
Figure 1	End-to-end research architecture
Figure 2	SPY price / return context
Figure 3	20D volatility + regime timeline
Figure 4	Walk-forward + purge diagram
Figure 5	Overall OOS MAE comparison
Figure 6	LowVol vs HighVol performance
Table 1	Dataset summary
Table 2	Feature definitions
Table 3	Model trade-offs/config summary
Table 4	Walk-forward design
Table 5	LightGBM overall/regime results
Table 6	Multi-model OOS comparison
Table 7	Cross-model regime comparison

Section Ownership

Section	Main owner
I–II	Tech Leader
III	Tech Leader + Model
IV.1–IV.5	Model + Data
IV.6	Pipeline

Section	Main owner
V	Data
VI.1–VI.6	Model
VI.7	Model
VII	Model
VIII	Tech Leader
IX	Tech Leader
Appendix / Reproduction	Pipeline

Writing Standard

Mỗi subsection nên đi theo:

Question → Method → Evidence → Interpretation

Ví dụ:

- **Question:** Does LightGBM beat the zero baseline?
- **Method:** Same genuine OOS dates and same metric implementation.
- **Evidence:** MAE, correlation, hit rate.
- **Interpretation:** State whether incremental predictive value was established.

Không viết generic claims như:

| LightGBM is strong because it is a powerful algorithm.

Chỉ claim điều experiment hoặc literature thực sự support.

Target Length

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Target main report: khoảng 22–28 trang, chưa tính Appendix.

Final Frozen Structure

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